

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP20016
Subject Title	Electromagnetism and Waves
Credit Value	3
Level	2
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	To introduce the basic concepts and laws in electromagnetism, and to formulate the laws in vector format. To provide a fundamental understanding of the principles of waves in light waves.
Intended Learning Outcomes	Upon completion of the subject, students will be able: (a) basic vectorial operations and partial differential equations formulation; (b) basic laws of electromagnetism in vector notation including Coulomb's law, Gauss' law, Biot-Savart law, Ampere's circuital law, and Maxwell's equations; (c) solve problems related to electromagnetism using different analytical techniques such as image method, boundary value approaches, the symmetric conditions and energy method; (d) possess a rigorous understanding of basic wave phenomena; (e) solve practical problems on waves; and (f) recognize the importance of waves in other branches of physics.
Subject Synopsis/ Indicative Syllabus	Vectors and vector fields: vector algebra; vector calculus & integrals, gradient, divergence, Stokes operation, the Laplacian, Curl, coordinate systems. Partial differential equations (PDE): first order PDE, Laplace equation, wave equation, initial and boundary value problems. Electrostatic calculations: investigation of vector electric fields generated by stationary charge distributions using Maxwell's equations. Using boundary conditions, image methods, complex analysis, solution of Poisson's equations etc for the investigations. Time-independent Maxwell equations: reviews the force laws of electrostatics and magnetostatics as vector field equations. Wave theory: harmonic waves, plane and spherical waves, principle of superposition, standing waves, beats, dispersion and group velocity. Mechanical waves: transverse waves on a string, longitudinal waves in a thin bar and in a fluid, sound waves, energy propagation, wave equations, d'Alembert's solution, boundary conditions. Light waves: electromagnetic waves, polarization, Poynting vector, intensity, light propagation in a medium.

Teaching/Learning Methodology	Lecture: The course materials will be explained. Concrete examples will be given to illustrate difficult concepts. Students are encouraged to ask questions and participate more in lectures. Assignment sets will be given to assess the learning progress of students. Tutorial: Students will work on problem sets in the tutorials, which provide them opportunities to apply the knowledge gained in lectures.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	1. Continuous Assignments	40	√	√	√	√	√	√
	2. Examination	60	√	√	√	√	√	√
	Total	100						
	Continuous assessment includes assignments and tests which aim at checking the progress of students throughout the course, assisting them in self-monitoring their performances. The examination will be used to assess the knowledge acquired by the students, as well as to determine the extent in which they have achieved the intended learning outcomes.							
Student Study Effort Expected	Class contact:							
	▪ Lecture & Tutorial					39 Hrs.		
	Other student study effort:							
	▪ Reading and self-study					81 Hrs.		
	Total student study effort					120 Hrs.		
Reading List and References	R. Fitzpatrick, “Maxwell’s equations and the principles of Electromagnetism”, Infinity Science Press LLC, 2008. W.H. Hayt, and J.A. Buck, “Engineering Electromagnetics”, 7th Edition, McGraw-Hill, 2006. G. King, “Vibrations and Waves”, Manchester Physics Series, Wiley, 2009. D. Halliday, R. Resnick and J. Walker, “Fundamentals of Physics”, 9th Edition, Wiley, 2010.							